Direction Matters: On the Implicit Regularization Effect of Stochastic Gradient Descent with Moderate Learning Rate

Jingfeng Wu, Difan Zou, Vladimir Braverman, Quanquan Gu

Johns Hopkins University & UCLA

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Overview

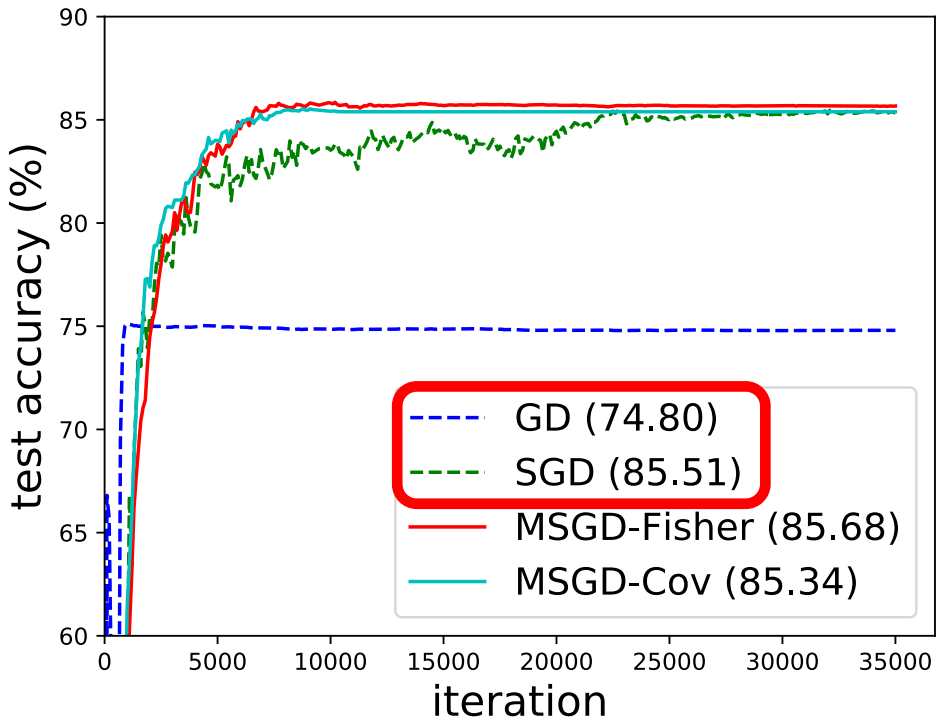
- Background
- SGD vs. GD: Different Convergence Directions
  - Small Learning Rate
  - Moderate Learning Rate
- SGD + Moderate LR is Good
- Proof Sketches

Implicit Regularization: SGD >> GD

Loss $$L_{\mathcal{S}}(w) = \frac{1}{n}\sum_{i=1}^{n} \ell_i(w)$$

GD $$w \leftarrow w - \eta \nabla L_{\mathcal{S}}(w)$$

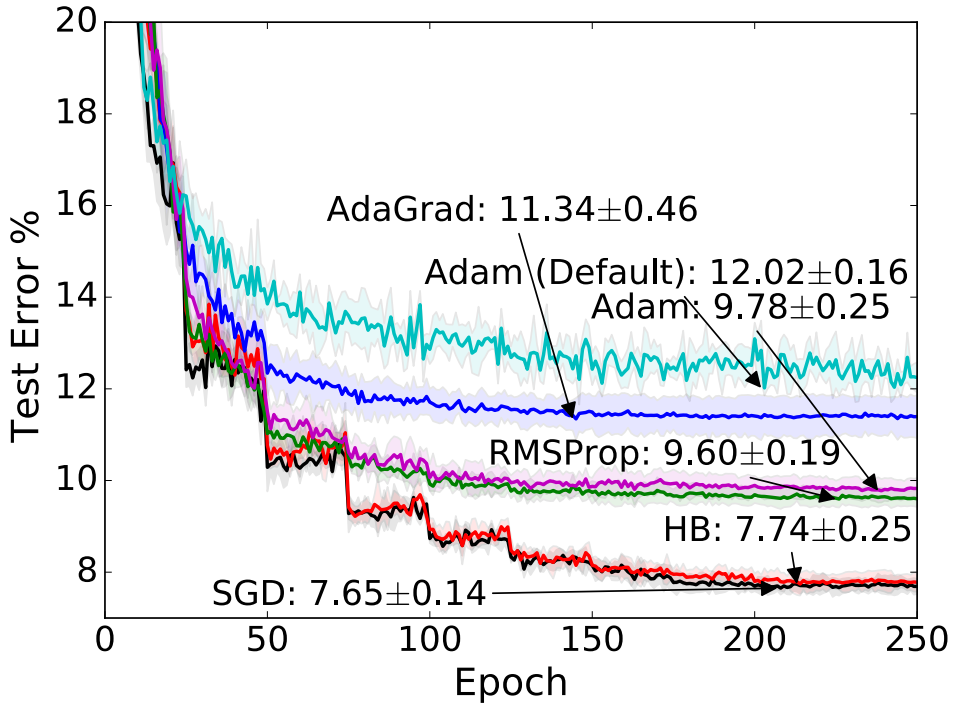
SGD $$w \leftarrow w - \eta \nabla \ell_k(w)$$

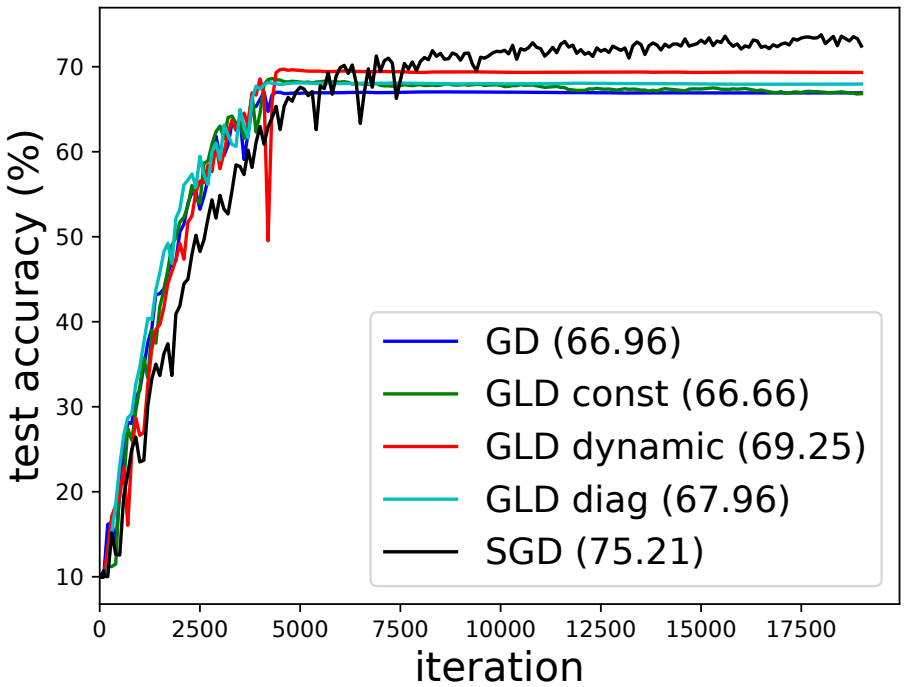


CIFAR-10, ResNet-18, w/o weight decay, w/o data augmentation

Wu, Jingfeng, et al. "On the Noisy Gradient Descent that Generalizes as SGD." ICML 2020.

Two More Figures about SGD (Less Relevant)

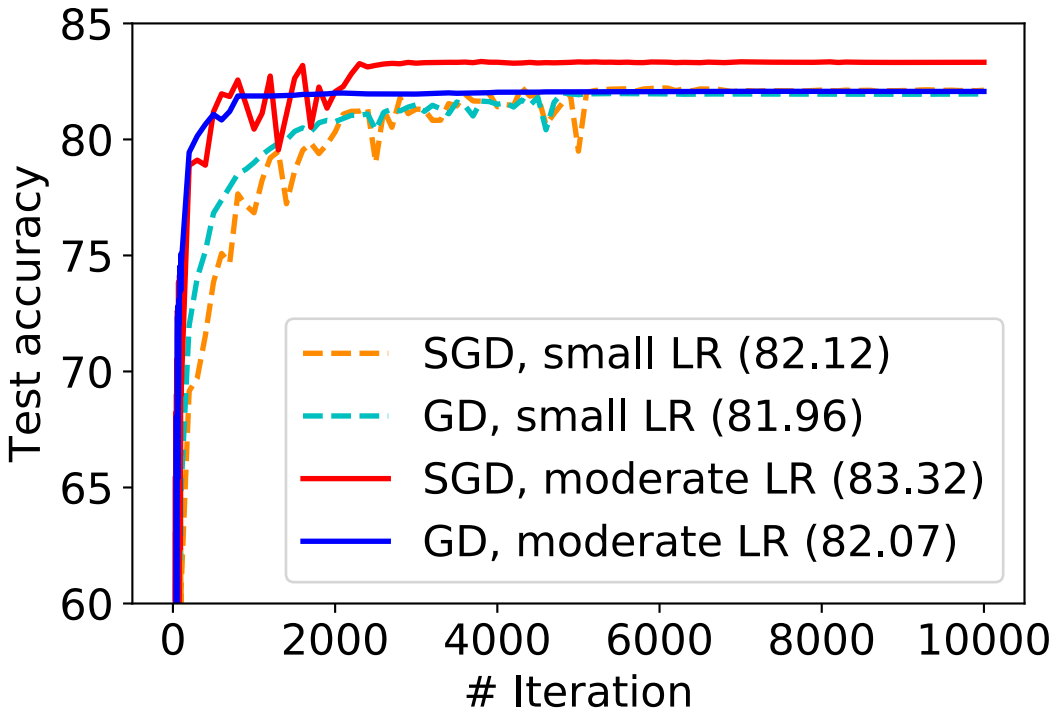




Wilson, Ashia C., et al. "The marginal value of adaptive gradient methods in machine learning." NIPS 2017.
Zhu, Zhanxing, et al. "The Anisotropic Noise in Stochastic Gradient Descent: Its Behavior of Escaping from Sharp Minima and Regularization Effects." ICML 2019.

SGD vs. GD: Learning Rate Matters!

	Small LR	Moderate LR
GD	☹	☹
SGD	☹	☺



Q1: Small LR, SGD ≈ GD?

Q2: Moderate LR, SGD >> GD?

Q3: GD is bad anyhow?

In Theory, SGD ≈ GD or SGD ≠ GD ??

GD $w \leftarrow w - \eta \nabla L_{\mathcal{S}}(w)$

SGD $w \leftarrow w - \eta \nabla L_{\mathcal{S}}(w) + \boxed{\eta \left(\nabla L_{\mathcal{S}}(w) - \nabla \ell_k(w)\right)}$

Unbiased noise (scales with η)

Theory disagrees with practice ☹

- "Easy" to prove SGD ≈ GD by concentration <= e.g., small LR
- "Hard" to prove an inverse result <= no concentration!

Small Learning Rate: SGD ≈ GD

GD $w \leftarrow w - \eta \nabla L_{\mathcal{S}}(w)$

$\eta = dt \to 0$

Gradient Flow (GF) $\mathrm{d}w = -\nabla L_{\mathcal{S}}(w)\mathrm{d}t$

Stochastic Modified Equation (SME) $\mathrm{d}w = -\nabla L_{\mathcal{S}}(w)\mathrm{d}t + \boxed{\sqrt{\eta}\Sigma(w)^{\frac{1}{2}}\mathrm{d}B_t}$

Higher order term

$\eta = dt \to 0$

SGD $w \leftarrow w - \eta \nabla L_{\mathcal{S}}(w) + \boxed{\eta\left(\nabla L_{\mathcal{S}}(w) - \nabla \ell_k(w)\right)}$

$$\begin{cases} \mathrm{Var}[\epsilon] = \sigma^2 \\ \mathrm{Var}[\epsilon_1 + \cdots + \epsilon_\eta] = \eta\sigma^2 \sim \mathcal{O}\left(\eta^2\right) \end{cases} \Rightarrow \sigma = \mathcal{O}\left(\sqrt{\eta}\right)$$

Effects of Non-Small Learning Rate

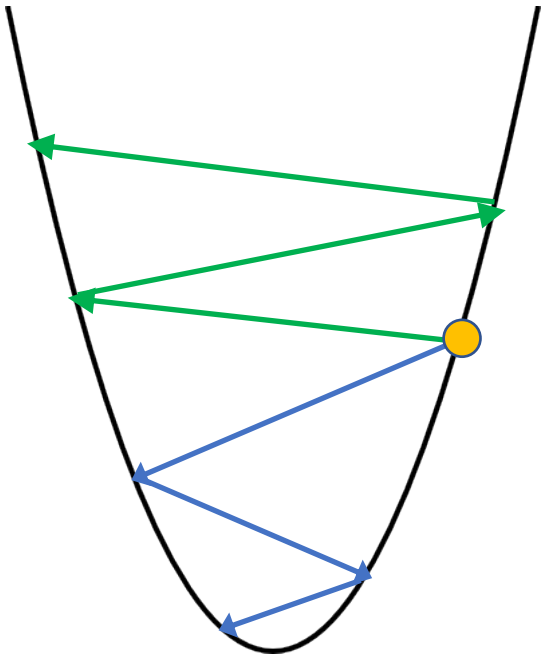
GD $\quad w \leftarrow w - \eta \nabla L_{\mathcal{S}}(w)$

SGD $\quad w \leftarrow w - \eta \nabla \ell_k(w)$

$$L_{\mathcal{S}}(w) = \frac{1}{n} \sum_{i=1}^{n} \ell_i(w)$$

Always smooth!

One of them can be non-smooth



$\eta \geq \frac{2}{\|H\|_2}$

$\eta < \frac{2}{\|H\|_2}$

$L(w) = 0.5\, w^{\top} H w$

$w_{k+1} = (I - \eta H) \cdot w_k$

SGD + moderate and annealing LR

- Phase 1: moderate LR => fits smooth losses
- Phase 2: small LR => fits non-smooth losses

A 2-D Example

$$\ell_1(w) = 0.5\, w^\top H_1 w, \quad H_1 = diag(2\kappa, 0)$$

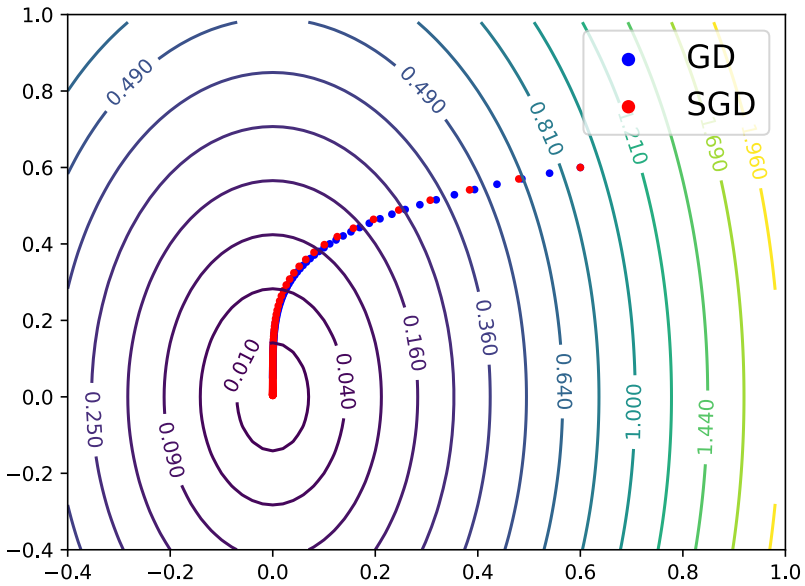
$$\ell_2(w) = 0.5\, w^\top H_2 w, \quad H_2 = diag(0, 2)$$

$$L(w) = 0.5\, w^\top H w, \quad H = diag(\kappa, 1)$$

$\kappa > 2$

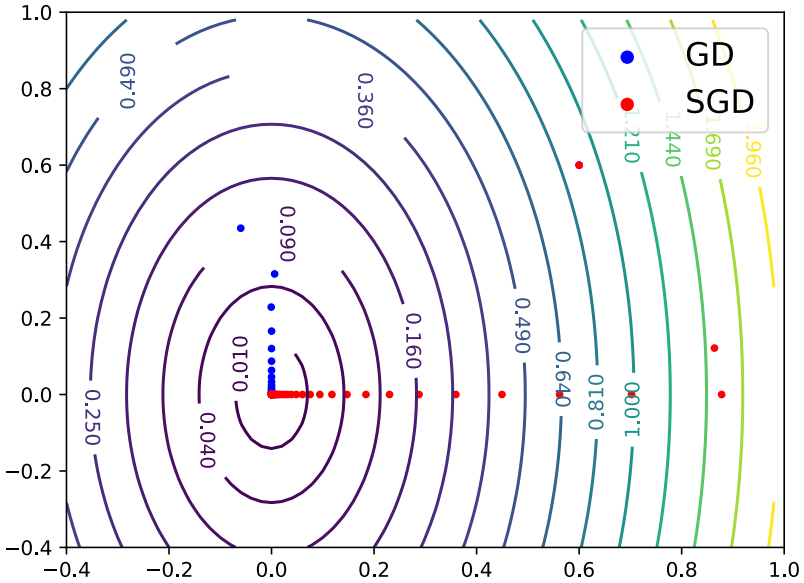
Small LR

$$\eta_t = {}^{0.1}/_{\kappa}, t = 1, \dots, T$$



Moderate LR

$$\eta_t = \begin{cases} {}^{1.1}/_{\kappa}, t = 1, \dots, T_1 \\ {}^{0.1}/_{\kappa}, t = T_1 + 1, \dots, T_2 \end{cases}$$



Same limits

Different convergence directions

A High Dimensional Linear Regression

Setups

- Test data $x = \zeta \cdot \xi \in \mathbb{R}^d$, where $\begin{cases} \zeta \in (0,1] \\ \xi \sim \mathcal{U}(S^{d-1}) \end{cases}$
- $\ell(x; w) = (w - w_*)^\top x^\top x (w - w_*)$
- Training data $\mathrm{X} = (x_1, \dots, x_n)$, $i.i.d.$, $d \gg n$

WOLG

- Let $\lambda_i = \|x_i\|_2^2 \in (0,1]$
- Assume $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$
- Let P be the projection onto the column space of X
- $P_\perp = I - P$

Theorem 0:

- There are multiple minima for $L_S(w) = \frac{1}{n}(w - w_*)^\top X X^\top (w - w_*)$
- The iterates of gradient methods belong to a hypothesis class $\mathcal{H}_S = \{w : P_\perp w = P_\perp w_0\}$
- If gradient methods find a global minima, then it is the one closest to initialization

Remark: this is also known as “minimal-norm solution” since the initialization is usually zero

A High Dimensional Linear Regression

Theorem 1 (informal): Consider SGD with moderate LR,

$$\eta_t = \begin{cases} \eta \in \left({}^{1}/_{\lambda_1} + o(1), {}^{1}/_{\lambda_2} - o(1) \right), & t = 1, \dots, T_1 \\ o(1), & t = T_1 + 1, \dots, T_2 \end{cases}$$

then

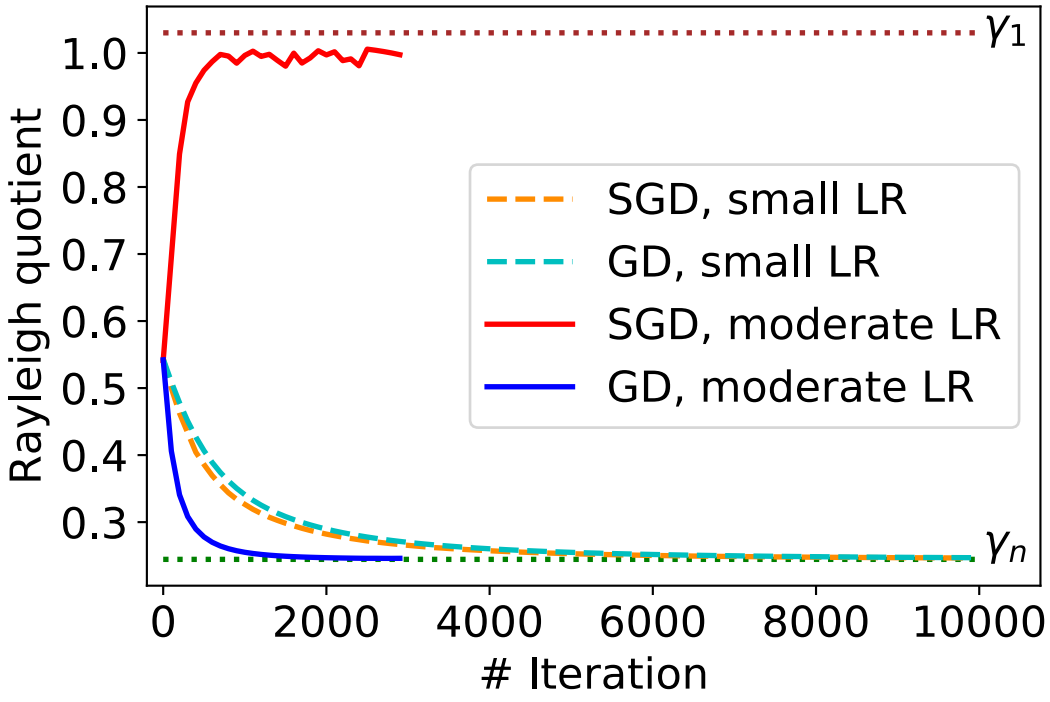
$$\frac{P(w - w_*)}{\|P(w - w_*)\|_2} \to v_1 \pm o(1)$$

Theorem 2 (informal): Consider GD with moderate or small LR,

$$\eta_t \in \left(0, {}^{n}/_{2\lambda_2} - o(1) \right), \qquad t = 1, \dots, T_2$$

then

$$\frac{P(w - w_*)}{\|P(w - w_*)\|_2} \to v_n \pm o(1)$$



Rayleigh quotient: $R(XX^\top, u) = \frac{u^\top XX^\top u}{u^\top u}$

Remark: v_1 (v_n) is the largest (smallest) eigen vector of $XX^\top$

Convergence Direction Matters

$$L_{\mathcal{D}}(w^{\mathrm{alg}}) - \inf_{w} L_{\mathcal{D}}(w) = \underbrace{L_{\mathcal{D}}(w^{\mathrm{alg}}) - \inf_{w' \in \mathcal{H}_{\mathcal{S}}} L_{\mathcal{D}}(w')}_{\Delta(w^{\mathrm{alg}}),\ \text{estimation error}} + \underbrace{\inf_{w' \in \mathcal{H}_{\mathcal{S}}} L_{\mathcal{D}}(w') - \inf_{w} L_{\mathcal{D}}(w)}_{\text{approximation error}}$$

determined by the algorithms and hyperparameters

intrinsic error, not improvable

- α-level set: $\mathcal{W}_\alpha = \{w \in \mathcal{H}_{\mathcal{S}} : L_{\mathcal{S}}(w) = \alpha\}$
- Optimal estimation error within a level set: $\Delta_\alpha^* = \min_{w \in \mathcal{W}_\alpha} \Delta(w)$

Theorem 3:

- For SGD with moderate LR, $\Delta\left(w^{sgd}\right) < \left(1 + o(1)\right) \cdot \Delta_\alpha^*$
- For GD with moderate or small LR, $\Delta\left(w^{gd}\right) > \left({}^{\gamma_1}/_{\gamma_n} - o(1)\right) \cdot \Delta_\alpha^*$

Remark: γ_1 (γ_n) is the largest (smallest) eigenvalue of $XX^\top$

Proof Sketch of Theorem 2

- $L_{\mathcal{S}}(w) = \frac{1}{n} w^\top X X^\top w$
- GD: $w_{t+1} = \left(I - \frac{2\eta}{n} X X^\top\right) w_t \quad \Rightarrow \quad w_t = \left(I - \frac{2\eta}{n} X X^\top\right)^t w_0 \quad \Rightarrow \quad u_t = \left(I - \frac{2\eta}{n} \Gamma\right)^t u_0$
- Moderate/small LR: $0 \leq \left(1 - \frac{2\eta\gamma_1}{n}\right)^t \ll \cdots \ll \left(1 - \frac{2\eta\gamma_n}{n}\right)^t \ll 1$
- Hence GD converges slower along the small eigenvalue directions

Proof Sketch of Theorem 1

- $\ell_i(w) = w^\top x_i x_i^\top w$
- SGD in one epoch: $w_{k,j+1} = \left(I - 2\eta x_{\pi(j)} x_{\pi(j)}^\top\right) w_{k,j} \quad \Rightarrow \quad w_{k+1} = \prod_{j=1}^{n}\left(I - 2\eta x_{\pi(j)} x_{\pi(j)}^\top\right) w_k$
- Bounding the spectrum of $\prod_{j=1}^{n}\left(I - 2\eta x_{\pi(j)} x_{\pi(j)}^\top\right)$ as projected onto $X_{-1} = (x_2, \dots, x_n)$ and its complement X/X_{-1}
- Bounding w_k as projected onto X_{-1} and X/X_{-1}
- And play with inequalities…

Remark: with moderate LR, concentration of matrix products turns out to be vacuous

Henriksen, Amelia, and Rachel Ward. "Concentration inequalities for random matrix products." Linear Algebra and its Applications 594 (2020): 81-94.

Take Home

- SGD + moderate LR: converges along large eigenvalue directions
- GD or SGD + small LR: converge along small eigenvalue directions
- The former directional bias benefits generalization
- The analysis is “anti-concentration”

Get the paper ->